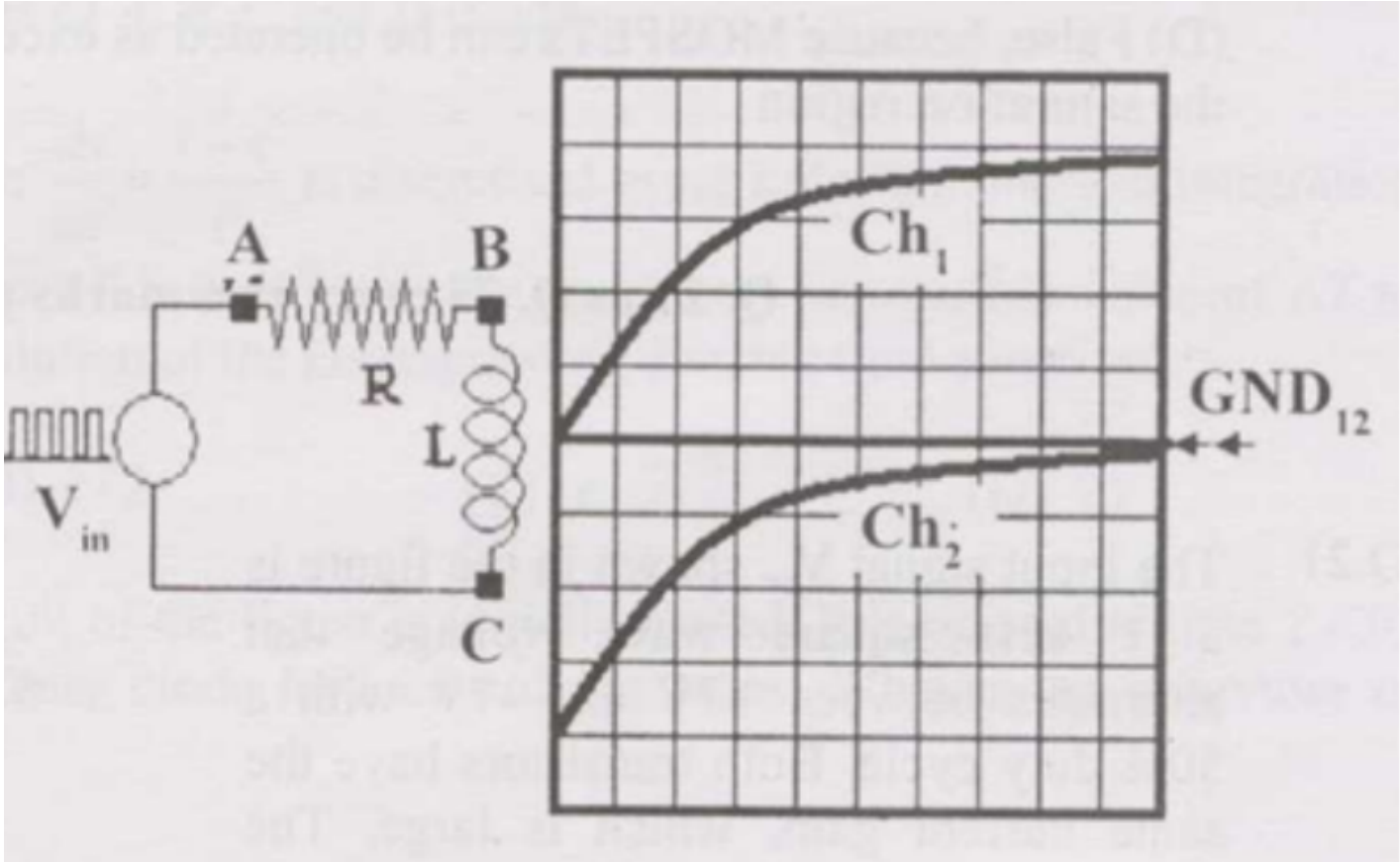
1.1.31

Sai Sreevallabh - EE25BTECH11031

Question

The probes of a non-isolated, two-channel oscilloscope are clipped to points A, B and C in the circuit as shown. V_{in} is a square wave of a suitable low frequency. The display on Ch_1 and Ch_2 are as shown on the right. Then the Signal and Ground probes S_1, G_1 and S_2, G_2 of Ch_1 and Ch_2 respectively are connected to points

Figure



Options

a) A,B,C,A

b) A,B,C,B

c) C,B,A,B

d) B,A,B,C

Theoretical Solution

The DSO plots voltage vs time graphs of the signal with respect to the chosen ground, as per the clipping of the probes.

A non-isolated oscilloscope has the ground pins of both the channel permanently connected to each other and they are also connected to the Earth ground of the wall outlet. If the ground probes are not connected to the same node, then the output will be flawed.
For the RL Circuit given in the figure, let the current be i. The circuit is described by the equation

$$V_{in} = iR + L\frac{di}{dt}$$

$$\frac{di}{V_{in} - iR} = \frac{dt}{L}$$

Theoretical Solution

Integrating both sides from assuming initial current $i(0) = 0$

$$\int_0^{i(t)} \frac{1}{V_{in} - i'R}\, di' = \int_0^t \frac{1}{L}\, dt'$$

Solving the integral, we get

$$i(t) = \frac{V_{in}}{R}\left(1 - e^{-\frac{R}{L}t}\right)$$
$$\implies V_R(t) = V_{in}\left(1 - e^{-\frac{R}{L}t}\right)$$

The waveform of Channel-1 of the DSO corresponds to this above equation. Hence, the signal probe (S_1) of the first channel is connected to point A and ground probe (G_1) is connected to point B.

Theoretical Solution

Using $V_L = V_{in} - V_R$ we get

$$V_L = V_{in} e^{-\frac{R}{L}t}$$

This expression corresponds to the waveform which begins at V_{in} at $t = 0$ and exponentially decays, with the X-axis as an asymptote. The waveform corresponding to Channel-2 is the negative of this exponentially decaying curve.

Hence, we can conclude that S_2 is connected to C and G_2 is connected to B. This is consistent with the fact that the ground probes of both the channels should be connected to the same node.

$$\boxed{Answer : B)\ A, B, C, B}$$

Simulation Code (Python)

```python
import numpy as np
import matplotlib.pyplot as plt

Vin = 5.0
R = 1000.0
L = 0.01
T = L/R #time constant

t = np.linspace(0, 5*T, 1000)

Vr = Vin*(1 - np.exp(-t/T)) #Voltage across resistor: ch-1
Vl = Vin*np.exp(-t/T) #voltage across L: negative of ch-2

plt.plot(t, Vr, color = 'red', label = "Ch-1")
plt.plot(t, -Vl, color = 'blue', label = "Ch-2")
plt.axhline(y=5.0, color='gray', linestyle='--')
```

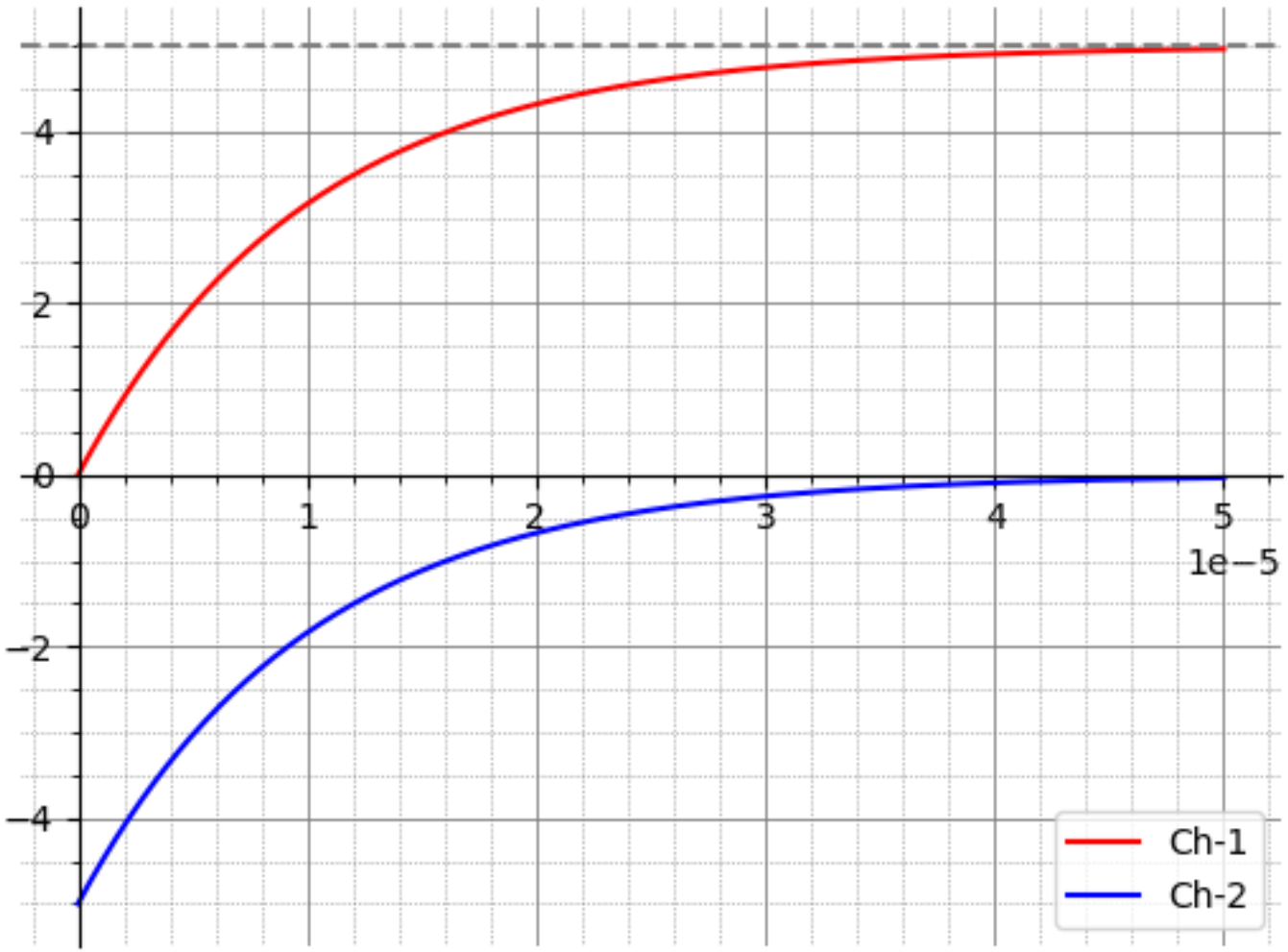
Simulation Code (Python)

```
ax = plt.gca()
ax.spines['top'].set_color('none')
ax.spines['bottom'].set_position('zero')
ax.spines['right'].set_color('none')
ax.spines['left'].set_position('zero')
plt.legend(loc='best')


plt.minorticks_on()
plt.grid(which='major', linestyle='-', linewidth='0.6', color='
    gray')
plt.grid(which='minor', linestyle=':', linewidth='0.5', color='
    gray')

plt.show()
plt.savefig("Figs/plot.png")
```

Simulation Plot



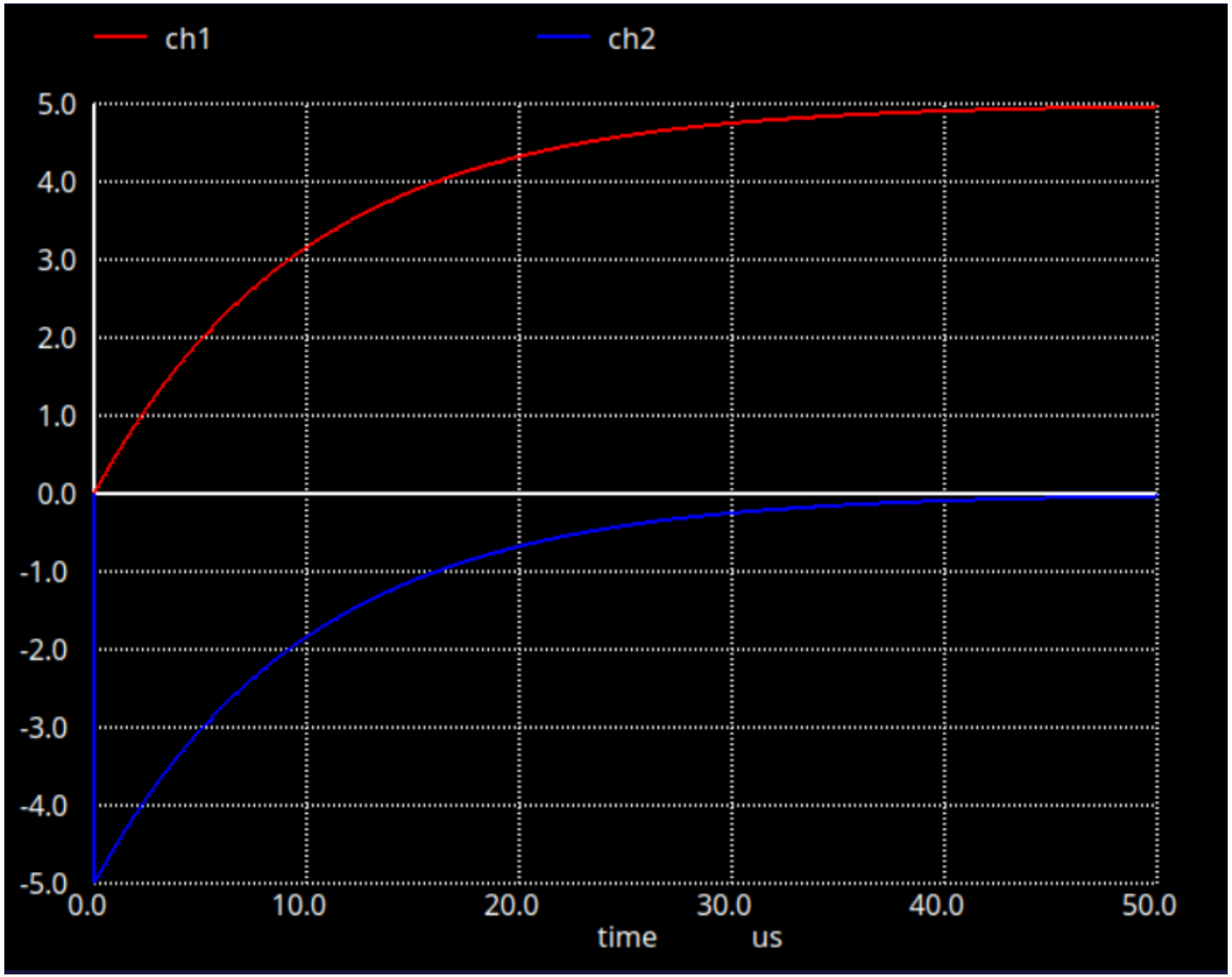
Simulation Code (NGSPICE)

```
*1.1.31

V1 in 0 PULSE(0 5 0 1ns 1ns 0.5s 1s)
R1 in x 1k
L1 x 0 10m

.control
tran 0.01us 50us

let Ch1 = v(in) - v(x)
let Ch2 = -v(x)

plot Ch1 Ch2
.endc
```

Simulation Plot NGSPICE